

Novel composite phase change materials supported by oriented carbon fibers for solar thermal energy conversion and storage

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HIGHLIGHTS

- Novel composite phase change materials supported by oriented carbon fibers are proposed
- High vertical thermal conductivity of $5.84 \text{ W}\cdot\text{K}^{-1}\cdot\text{m}^{-1}$ and low horizontal thermal conductivity of $1.34 \text{ W}\cdot\text{K}^{-1}\cdot\text{m}^{-1}$ are obtained
- High solar-to-thermal efficiencies of $87.54\% \sim 95.08\%$ are obtained at $1\sim 3 \text{ kW}\cdot\text{m}^{-2}$
- The composite phase change material exhibits good stability for long-term utilization

ARTICLE INFO

Keywords:

Oriented carbon fiber
Composite phase change material
Anisotropic thermal conductivity
Solar-to-thermal efficiency

ABSTRACT

Phase change materials (PCMs) have aroused significant interest as promising materials for solar thermal energy conversion and storage. However, the long-standing shortcomings of liquid leakage, low thermal conductivity, and weak solar absorptance limit their practical applications. Herein, novel composite phase change materials (CPCMs) with anisotropic heat conduction are manufactured by mixing continuous carbon fibers (CFs) and palmitic acid (PA)/olefin block copolymer (OBC) mixtures using pressure induction and vacuum treatment. Because the oriented CFs in the vertical direction can offer heat transfer channels inside the composites, the vertical and horizontal thermal conductivities of the CPCMs are $5.84 \text{ W}\cdot\text{K}^{-1}\cdot\text{m}^{-1}$ and $1.34 \text{ W}\cdot\text{K}^{-1}\cdot\text{m}^{-1}$, respectively, resulting in a relatively high anisotropic degree of 4.36. Furthermore, to improve the absorption of solar radiation for the composite, carbon black is applied to the upper surface of the CPCMs, achieving a high total solar absorptance of 0.966. Due to the combination of the high solar absorptance of the carbon black and the high vertical thermal conductivity within the composites, the CPCMs exhibit outstanding solar-to-thermal efficiencies of $87.54\% \sim 95.08\%$ at $1\sim 3 \text{ kW}\cdot\text{m}^{-2}$. In addition, stability testing also confirms that the CPCMs have excellent leakage-proof properties, thermal stability, and cyclic stability for long-term utilization. This work can provide an efficient route to synthesize high-performance CPCMs for solar thermal applications.

1. Introduction

Solar thermal energy conversion and storage technology is essential for the effective utilization of abundant solar energy for industrial heating, hot water supply, and other heating-related applications [1–3]. However, the intermittent and erratic nature of solar irradiation seriously limits the extensive harnessing of solar energy. Phase change materials (PCMs) have developed into crucial ingredients for solar thermal energy harvesting due to their isothermal phase change

properties and high heat storage capacity, thus overcoming the discontinuous properties of solar irradiation [4]. However, most PCMs are constrained by their poor shape stability, low thermal conductivity, and weak solar absorptance, resulting in slow thermal response and low solar-to-thermal efficiency [5]. Therefore, developing functional composite phase change materials (CPCMs), which are made by chemical modification or the addition of functional fillers, is critical to alleviate these drawbacks and enhance the large-scale applications of PCMs [6].

In recent years, to resolve the leakage of molten PCMs during melting

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and crystallization processes, extensive research has been focused on synthesizing form-stable CPCMs by embedding PCMs into the polymer [7]. The polymer serves as an adsorption and supporting material to enhance the leakage-proof properties of the PCMs. Some polymers, containing olefin block copolymer (OBC) [8], polyethylene [9], thermoplastic elastomer [10], etc., have excellent compatibility with commonly used PCMs. However, because the solar absorptances and thermal conductivities of these polymers are equivalent to those of the PCMs, most CPCMs still suffer from weak solar absorptance and poor thermal conductivity, which can be detrimental to efficient heat absorption and rapid heat spreading inside the composites [11].

To tackle these issues, extensive efforts have been made to add various highly thermal conductive carbon-based nano-additives, including graphene [12], carbon nanotubes [13], graphite [14], etc., into PCMs, which can achieve considerably increased thermal conductivity and micro-convection at the nanoscale [15]. For example, Chen et al. [16] designed CPCMs by embedding graphene oxide into paraffin wax, and their thermal conductivity can reach $0.46 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and their total solar absorptance can be up to 0.952. Mehrali et al. [17] prepared graphene-based CPCMs, and the results show that the thermal conductivity of the obtained CPCMs is $1.93 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and their total solar absorptance can reach 0.86. Zhang et al. [18] fabricated CPCMs supported by Ag-graphene nanoparticles, showing that the thermal conductivity and total solar absorptance of the CPCMs are $0.41 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and 0.92, respectively. Qian et al. [19] proposed CPCMs with single-walled carbon nanotubes, demonstrating that their thermal conductivity can reach $2.7 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and their solar-to-thermal efficiency can reach 78% at $1 \text{ kW}\cdot\text{m}^{-2}$ (i.e., 1 sun). Zhu et al. [20] fabricated CPCMs by embedding graphene nano-additive into n-octadecane, and the results indicate that their thermal conductivity and solar-to-thermal efficiency under $1 \text{ kW}\cdot\text{m}^{-2}$ can reach $0.80 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and 70.0%, respectively. However, these nanoparticles still show inadequate and unstable thermal conductivity enhancements, which can be attributed to two reasons. One is that nanoparticles are distributed randomly inside the composites, failing to form continuous thermal transport pathways, leading to high interfacial thermal resistances between the PCMs and nanoparticles [21]. The other is that dispersed nanoparticles can aggregate and settle during the phase change process, which can hinder phonon transport and limit the heat transfer within the composites adversely [22].

Compared to the nanoparticles added within the composite, three-dimensional carbon materials, including graphene foams [23], carbon foams [24], carbon nanotube sponges [25], etc., are selected to construct cross-linked thermal networks, which can provide continuous heat transfer paths to accelerate the thermal response within the composites. For example, Aftab et al. [26] designed CPCMs by infiltrating polyethylene glycol into carbon nanotube sponges, showing that their thermal conductivity is $2.4 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and their solar-to-thermal efficiency can be up to around 80% under $1 \text{ kW}\cdot\text{m}^{-2}$. Zhang et al. [27] prepared CPCMs supported by 3D porous foam, and the results indicate that the thermal conductivity and solar-to-thermal efficiency of the CPCMs can reach $1.04 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and 86.68%, respectively. Yang et al. [28] designed graphene foam-based CPCMs, demonstrating that their thermal conductivity is $1.82 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which is 574% higher than that of the paraffin wax. Wei et al. [29] fabricated CPCMs by using carbon aerogels, and the results show that their thermal conductivity is $0.43 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and solar-to-thermal efficiency can be as high as 82% under $3 \text{ kW}\cdot\text{m}^{-2}$. Chen et al. [30] proposed CPCMs using the graphitized network, showing that the thermal conductivity of the CPCMs is up to $0.91 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ with increasing filler loadings, which is enhanced by 236% compared with that of the polyethylene glycol. Cui et al. [31] designed CPCMs with modified melamine foam, and the results show that their thermal conductivity can reach $0.59 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and their solar-to-thermal efficiency is as high as 79.36% under $1 \text{ kW}\cdot\text{m}^{-2}$. Shen et al. [32] proposed all lignin-based sponge-encapsulated CPCMs, showing that the thermal conductivity of the composite is 0.36

$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and the spectral absorptance can be as high as around 0.9 when the wavelength is in the range of 0.2–1.0 μm . Lv et al. [33] fabricated phase change aerogel using graphene and silver nanoparticles, demonstrating that its thermal conductivity is improved to $0.856 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and its total solar absorptance can exceed 0.90 when the wavelength is in the range of 0.25–4 μm . Zhang et al. [34] designed wood-based CPCMs, showing that the thermal conductivity can reach $0.23 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, and the solar-to-thermal efficiency can be boosted to 83% at $2 \text{ kW}\cdot\text{m}^{-2}$.

Although the thermal conductivities of the CPCMs with three-dimensional carbon materials are enhanced, these values still exhibit low and isotropic, which can result in more heat being transferred from the interior of the composite to the environment (i.e., horizontal), thus reducing the solar-to-thermal efficiency [35]. Therefore, to weaken the heat dissipation to the surroundings and strengthen the heat conduction from the surface to the interior (i.e., vertical), orientation control of the thermally conductive filler is essential to realize anisotropic thermal transport in different directions inside the composite.

Recently, carbon fibers (CFs) have been considered as promising fillers to design anisotropically conductive CPCMs [36]. The directional alignment of the CFs can accelerate the thermal transport inside the composite, thereby increasing the thermal conductivity in a specific direction within the composite [11]. Meanwhile, the thermal conductivities in the other directions can be weakened, thus reducing heat dissipation to the surroundings. For example, Sheng et al. [37] designed CPCMs with carbon fiber frameworks, showing that their vertical thermal conductivity can be improved to $0.77 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at 8.8 wt% CFs, which is 133% higher than their horizontal value. Ma et al. [38] proposed polymer composites supported by short CFs, demonstrating that the vertical and horizontal thermal conductivities at 13.0 vol% CFs are $2.84 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and $0.19 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, respectively. Wu et al. [39] fabricated epoxy composites using CF networks, indicating that their vertical and horizontal thermal conductivities at 17.48 vol% CFs are up to $6.20 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and $1.24 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, respectively. Hou et al. [40] designed polymer composites by utilizing directional freezing to form oriented and ordered CFs connected by short CFs in PDMS, showing that the vertical thermal conductivity of the composite at 10.8 vol% CFs can be improved to $3.50 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, which is 368% higher than that of the composite with random CFs.

Despite oriented short CFs within the composite can lead to anisotropic heat conduction, the vertical and horizontal thermal conductivities reported above are much lower than expected for CFs-based materials. This is ascribed to the fact that larger interfacial thermal resistance appears at the CFs-CFs interface and PCMs-CFs interface, which can diminish the thermal response within the composite, leading to a considerable decrease in thermal conductivity [41]. Hence, low anisotropic thermal conductivity limited by large interfacial thermal resistance within the CPCM remains a critical challenge.

In this paper, we proposed a simple method by mixing long and continuous CFs and the palmitic acid (PA)/olefin block copolymer (OBC) mixtures using pressure induction and vacuum treatment to fabricate anisotropically conductive CPCMs, which can form uninterrupted heat transfer paths and decrease the interfacial thermal resistances at the CFs-CFs interface and PCMs-CFs interface, contributing to boosting the anisotropic thermal conductivity of the composites. Then, the morphology and structures of the obtained CPCM were investigated. Furthermore, the thermal properties of the obtained CPCM were analyzed. Then, the leakage-proof properties of the obtained CPCM were evaluated. Then, the anisotropic thermal conductivity and heat transfer properties of the CPCM were examined. Finally, the solar-to-thermal energy conversion and storage performance was measured and assessed within a simulated sunlight testing system.

2. Materials and methods

2.1. Materials

Carbon fibers (TC-HC-600) were obtained from Shanxi Tiance New Materials Technology Co., Ltd. China, and their axial thermal conductivity is around $600 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$. Olefin block copolymer (OBC, INFUSE 9530) was obtained from Dow Chemical Company. The carbon black (99%) was provided by Yueyang Dongrun Chemical Co., Ltd. China. Palmitic acid (PA) was provided by Shanghai Macklin Biochemical CO., Ltd. China.

2.2. Synthesis of CPCMs with oriented CFs

The rolled carbon fibers (CFs) were sheared into several dispersed short CFs and then arranged in a certain direction on the metal plate to form oriented continuous CFs. Then, the PA and OBC were mixed in a specific mass fraction and then stirred at 170°C for 1 h to make the molten PA/OBC blend. Subsequently, the molten blend was poured into the CFs in the metal plate and brushed evenly to assure that the CFs and PA/OBC blend could be fully mixed. Then, the obtained PA/OBC/CF mixture was cooled down and divided into many slices. Then, these slices were placed in a mold and compressed at 30 MPa to obtain shaped slices. Finally, the shaped slices with a mold were put in a vacuum oven and then heated at 70°C for 1 h, and compressed again to obtain the CPCMs with oriented CFs (Fig. 1).

In the CPCMs, PA provided high latent heat of fusion and crystallization, OBC served as supporting materials to improve the leakage-proof performance, and CFs served as highly thermally conductive additives to form continuous pathways for boosting the thermal conductivity of the composites. Furthermore, the specific information on the PA and obtained CPCMs can be found in Table 1.

2.3. Characterization

The morphology of the CPCMs in the vertical and horizontal directions was studied by using a scanning electron microscope (SEM, JSM-IT300LA, JEOL). The crystalline structures of the pure PA, OBC, CF, and CPCMs were analyzed by X-ray diffractometer (XRD, D8 Advance, Bruker). The chemical compositions of the pure PA, OBC, CF, and CPCMs were measured by fourier transform infrared spectroscopy (FTIR, Nicolet iS50, Thermo). The latent heat of fusion and crystallization and phase change temperatures of the PA and CPCMs were tested

Table 1

Fractions and densities of the PA and CPCMs.

Sample	PA / wt%	OBC / wt%	CF / wt%	Density / $\text{kg} \cdot \text{m}^{-3}$
PA	100	0	0	987.2
CPCM 1	95	5	0	952.1
CPCM 2	90	10	0	945.0
CPCM 3	85	15	0	944.5
CPCM 4	80	20	0	925.4
CPCM 5	75	20	5	1040.1
CPCM 6	70	20	10	1063.7
CPCM 7	65	20	15	1114.7
CPCM 8	60	20	20	1141.2

using a differential scanning calorimeter (DSC 214, Netzsch). A hot disk thermal constant analyzer (DZDR-S, Nanjing Dazhan) was used to evaluate the vertical and horizontal thermal conductivities of the obtained composites. The spectral absorptances of the composites were performed on an ultraviolet-visible-near infrared spectrophotometer (UV-3600i Plus, Shimadzu). The thermal decomposition performance was evaluated by a thermogravimetric analyzer (TGA, TG 209 F3, Netzsch) at $30\text{--}600^\circ\text{C}$. The heat transfer characteristics and leakage-proof performance were tested by putting the samples on a heating disk at 80°C . An infrared camera (MobIR Air, Guide) was employed to record the temperature distributions of the composites.

2.4. Solar-to-thermal conversion and storage performance test

A xenon lamp (CEL-PF300-T8, CEAuLight) was employed to irradiate the surface of the prepared samples at irradiation intensities within $1\text{--}3 \text{ kW} \cdot \text{m}^{-2}$. The simulated irradiation intensities were measured by an optical power meter (CEL-NP2000-2, CEAuLight). The temperatures of the bottom centers of the samples were measured by a J-type thermocouple (accuracy: $\pm 0.1^\circ\text{C}$). The solar-to-thermal efficiency (η) of the composites could be given by Eq. (1). The relative uncertainty of η (U_η) was obtained to be 4.27% (Note S1).

$$\eta = \frac{m_c \cdot \Delta H_m}{I \cdot S_c \cdot \Delta t} \quad (1)$$

where m_c and S_c represent the mass and the irradiated area of the composites, respectively; ΔH_m represents the latent heat of fusion of the composites; I represents the intensity of irradiation of the xenon lamp; Δt represents the illumination time during the phase transition process, which equals to the difference between the beginning time and the end time of the phase change during the melting process.

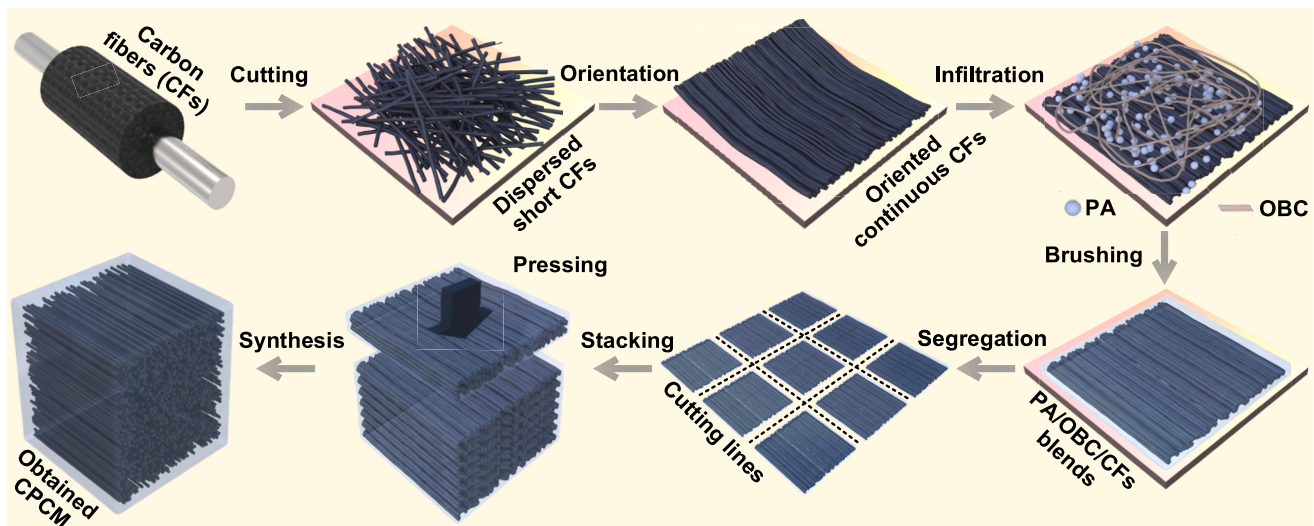


Fig. 1. Schematic diagram of the preparation process of the CPCMs.

3. Results and discussion

3.1. Morphology and structural properties of CPCMs

The morphology of the CPCM8 with 20 wt% CFs in the vertical and horizontal sections is presented in Fig. 2. From the CPCM8 in the vertical direction, it can be found that the surfaces of the CFs are covered well by the PA/OBC mixture (Fig. 2a-c). Meanwhile, the CFs form continuous heat transfer paths within the CPCMs, which can accelerate the thermal transport inside the CPCMs. Moreover, the CFs are distributed uniformly and aligned well, which can also be helpful to reduce the interfacial thermal resistance and boost the thermal conductivity of the composite uniformly (Fig. 2a-c). From the horizontal direction, the SEM images also clearly demonstrate that the CFs are relatively uniformly distributed in the PA/OBC mixture, and the diameters of the individual CFs can be seen to be around 5–8 μm (Fig. 2d-f). Besides, the CFs can also be aligned and dispersed well from both vertical and horizontal directions inside the CPCMs with different CF loadings (Fig. 2h).

Then, to analyze the structure properties of the CPCMs, firstly, the XRD patterns of PA, OBC, CF, and CPCM8 were measured by an X-ray diffractometer (Fig. 2g). It can be found that the PA exhibits two characteristic peaks from its XRD patterns, corresponding to the diffraction angles of 21.4° and 23.8° , respectively. For the OBC, the characteristic peak only occurs at around 21.5° . Furthermore, the XRD patterns of the CFs show two characteristic peaks, appearing at about 26.5° and 54.5° , respectively (Fig. 2g). For the CPCM8, it can be seen that the peaks at 21.5° , 23.8° , 26.5° , 54.5° are four main characteristic peaks of the composite, including all the peaks of the PA, OBC, CF (Fig. 2g). Besides, the peak intensities of the PA, OBC, and CF in the composite exhibit a notable reduction compared with those of their individual phases in the entire range, but there are no new characteristic peaks found in the CPCMs. Furthermore, the Fourier transform infrared spectroscopy (FTIR) spectra of the PA, OBC, CF, and CPCM8 were measured to further investigate the chemical composition of the composite (Fig. S1). It can be found that there are two major characteristic peaks at around 2915

cm^{-1} and 2850 cm^{-1} , which are ascribed to the $-\text{CH}_2$ asymmetrical stretching vibration and $-\text{CH}_2$ symmetrical, respectively [8]. Moreover, as presented in the spectrum of OBC, the peaks around 1480 cm^{-1} and 720 cm^{-1} are attributed to $-\text{CH}_2$ rocking [42]. For the spectrum of the CF, it can be found that almost no absorption peaks appear in the spectrum range from 4000 cm^{-1} to 400 cm^{-1} due to the lack of functional groups. In addition, the spectrum of the CPCM8 presents all the peaks of PA, OBC, and CF, showing that the main peaks of the composite remain unchanged and no other new peaks appear. These results demonstrate that the CPCM is only a physical combination of the PA, OBC, and CF, and no chemical interaction appears.

3.2. Thermal properties of CPCMs

To assess the thermal properties of the CPCMs, the DSC analysis of the melting and crystallization processes of the PA and different CPCMs was performed (Fig. 3, Table S1). It can be observed that the peak shapes of the composites are similar to those of the PA, with one endothermic peak after heating and one exothermic peak after cooling, which can indicate that the CPCMs exhibit similar thermal properties to pure PA, particularly with regard to the phase change temperature. As shown in Fig. 3a-b, the onset temperatures of the melting (T_m) and crystallization (T_c) of the PA are 63.5°C and 58.2°C , respectively. After the addition of OBC and CFs within the PA, both the T_m and T_c of the composites show a slight reduction compared with those of the PA, which can reach $60.0\text{--}62.7^\circ\text{C}$ and $51.6\text{--}56.0^\circ\text{C}$, respectively. The results confirm that the phase change temperature of the PA is rarely affected by the CFs and OBC.

Furthermore, it can be observed in Fig. 3c that the experimental latent heat of fusion (ΔH_m) and latent heat of crystallization (ΔH_c) decrease proportionally with decreasing PA loadings. This is attributed to the fact that the CFs, OBC, and PA are just physically combined, and OBC and CFs cannot provide the latent heat of fusion and crystallization due to their own inability to phase transition. Hence, the latent heat of the composites only varies linearly with PA loading and can be obtained

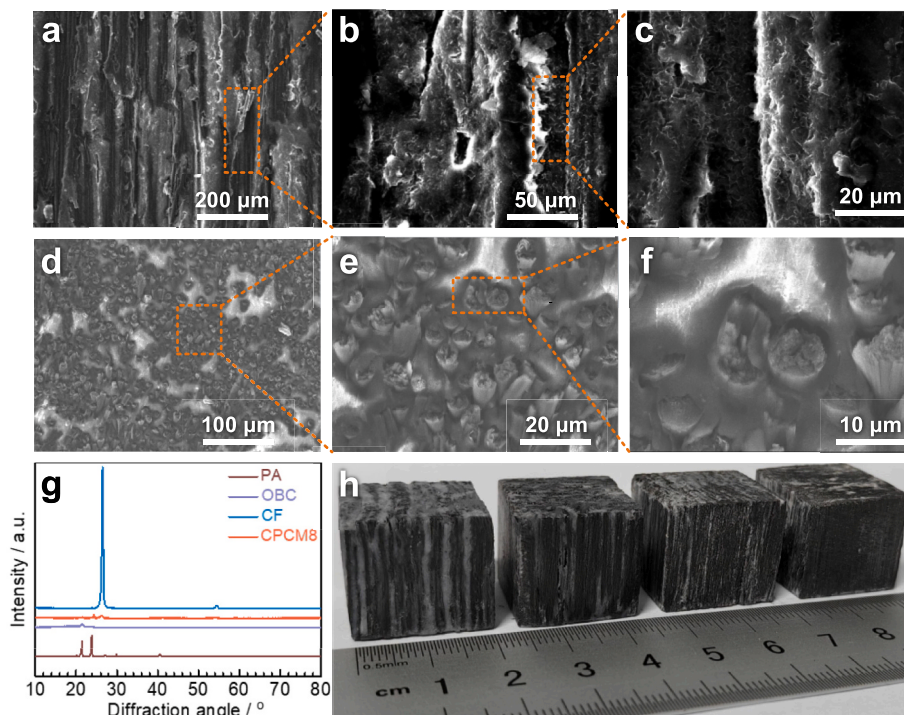


Fig. 2. Morphology and structural properties of CPCMs. (a)-(c) Morphology of the CPCM8 in the vertical section and its localized enlarged SEM images. (d)-(f) Morphology of the CPCM8 in the horizontal section and its localized enlarged SEM images. (g) XRD patterns of the PA, OBC, CF, and CPCM8. (h) Digital photographs of the CPCMs with different mass fractions of the CFs.

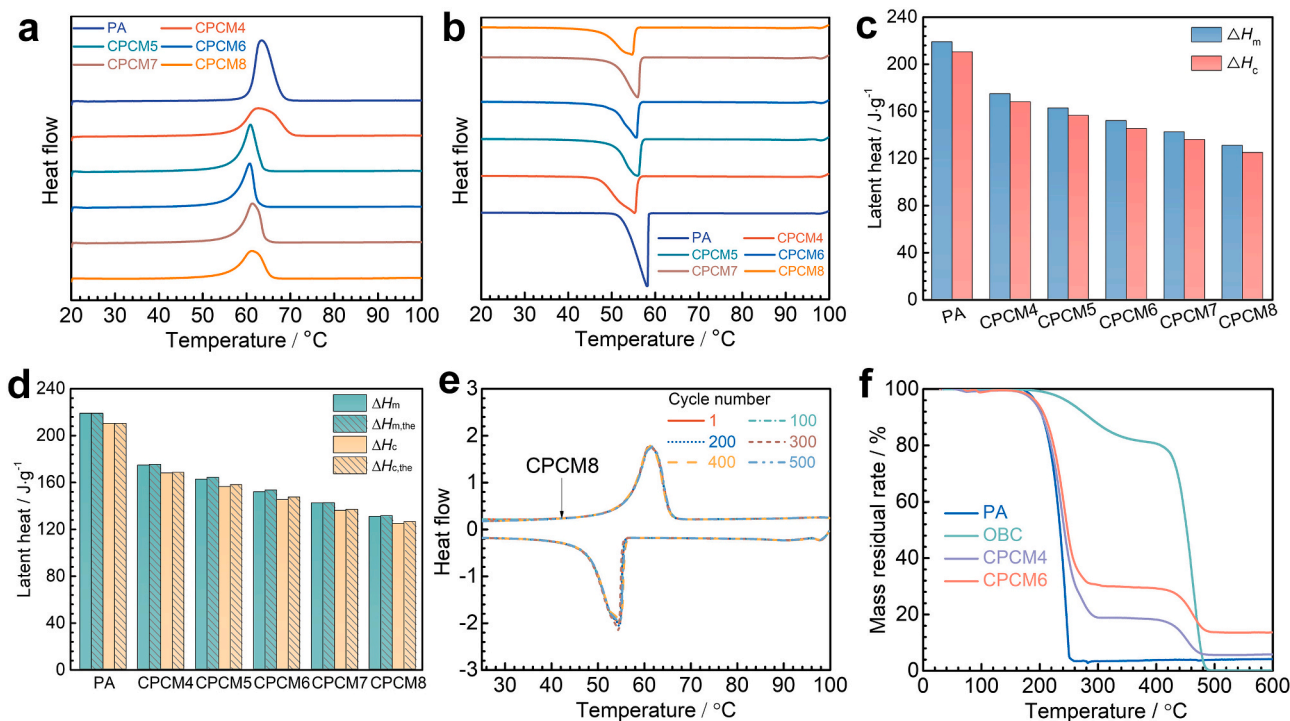


Fig. 3. Thermal properties of the pure PA and CPCMs. (a)–(b) DSC curves of the PA and CPCMs for melting and crystallization processes. (c) Comparisons of ΔH_m and ΔH_c of the PA and CPCMs. (d) Comparisons of $\Delta H_{m,the}$, ΔH_m , $\Delta H_{c,the}$ and ΔH_c of the PA and CPCMs. (e) Thermal cyclic stability of the CPC8 for 500 heating/cooling cycles. (f) Thermogravimetric curves of the CPCMs at 25–600 °C.

by Eq. (2).

$$\Delta H_{the} = \varphi_{PA} \cdot \Delta H_{PA} \quad (2)$$

where φ_{PA} and ΔH_{PA} represent the proportion and latent heat of the PA, respectively. It can be found that the calculated theoretical latent heat of fusion ($\Delta H_{m,the}$) and latent heat of crystallization ($\Delta H_{c,the}$) of the composites are consistent with the ΔH_m and ΔH_c obtained from the experiments, respectively (Fig. 3d). Besides, it is worth noting that the proportion of the PA in the composite should be reasonably considered to ensure a high thermal storage capacity. This is because lower PA loadings can lead to an obvious decrease in the thermal capacity of the composites, which is not conducive to thermal energy storage in practical applications. Hence, achieving the desired residual latent heat in the CPCMs depends significantly on the proportion of added CFs.

Then, the thermal cyclic stabilities of the CPCMs under long-term operation were examined. Fig. 3e presents the DSC curves of the CPC8 during 500 heating/cooling cycles at 25–100 °C. The result exhibits that there is no significant evolution in the DSC curves during the heating and cooling cycles, indicating that the CPCMs behave with long-term stability for solar thermal utilization.

Finally, the thermogravimetric analyzer was employed to assess the thermal stabilities of the PA, OBC, CPC4, and CPC6, as illustrated in Fig. 3f. The TGA curve of the PA starts to decompose obviously at about 190 °C and the decomposition process continues until the temperature reaches 250 °C. From the curve of the OBC, it shows a slight decomposition starting at 210 °C, and then a significant plateau can be observed at approximately 400 °C and there is an approximately 20% decrease in the mass residual rate within this temperature range. Finally, the decomposition of the OBC is completed at about 500 °C. In terms of the CPC4, its mass residual rate is reduced by around 80% within the temperature of 300–450 °C, which corresponds to the proportion of the OBC in the composite. Furthermore, it can be found for the CPC6 that its mass residual rate at 300–450 °C is around 30%, which is equal to the sum of the OBC and CF loadings. In addition, at the end of the experiment, the mass residual rate of the CPC6 is 10%, which is in

accordance with the proportion of CFs. These results confirm that the composites present excellent stability in the working range below 150 °C.

3.3. Leakage-proof properties of CPCMs

The leakage-proof property is a significant factor affecting the thermal properties and morphology of the CPCMs. Therefore, to assess the leakage-proof properties of the CPCMs, the pure PA and different CPCMs were placed on filter paper and exposed to a heating disk at 80 °C for 1 h, and their mass was recorded at 10-min intervals. It can be observed in Fig. 4a that the PA melted completely at 30 min and penetrated into the filter paper owing to the absence of leakage protection. In terms of the CPC3, the leaked PA also leaves a visible sign on the filter paper after 30 min, and its residual weight rate drops to 94% (obtained in Note S2) at the end of the experiment, which can demonstrate that the leakage of the composite still exists at OBC loading of 15 wt%. Furthermore, no significant sign is found in the CPC4 and the residual weight rate can reach 99% after heating for 60 min, indicating that 20 wt% OBC proves to be highly effective in preventing the molten PA from leaking out of the CPCMs. Thus, 20 wt% OBC is chosen as the anti-leakage agent for the composites. For the CPC5–CPC8 with 20 wt% OBC, it can be found that melted PA in the composites is hardly leaked during the entire process and their residual weight rates are almost unchanged (Fig. 4b).

Besides, the mass loss rates of CPC5–CPC8 diminish with increasing CF loading (Fig. 4c, Table S2), indicating that the addition of the CFs is helpful to weaken the flow characteristics of melted PA and improve the leakage-proof properties of the CPCMs. In general, the above findings confirm that the CPCMs with different CF loadings at 20 wt% OBC can keep stable and avoid leakage.

3.4. Anisotropic thermal conductivity and heat transfer characteristics of CPCMs

Thermal conductivity is essential to strengthen heat transfer within

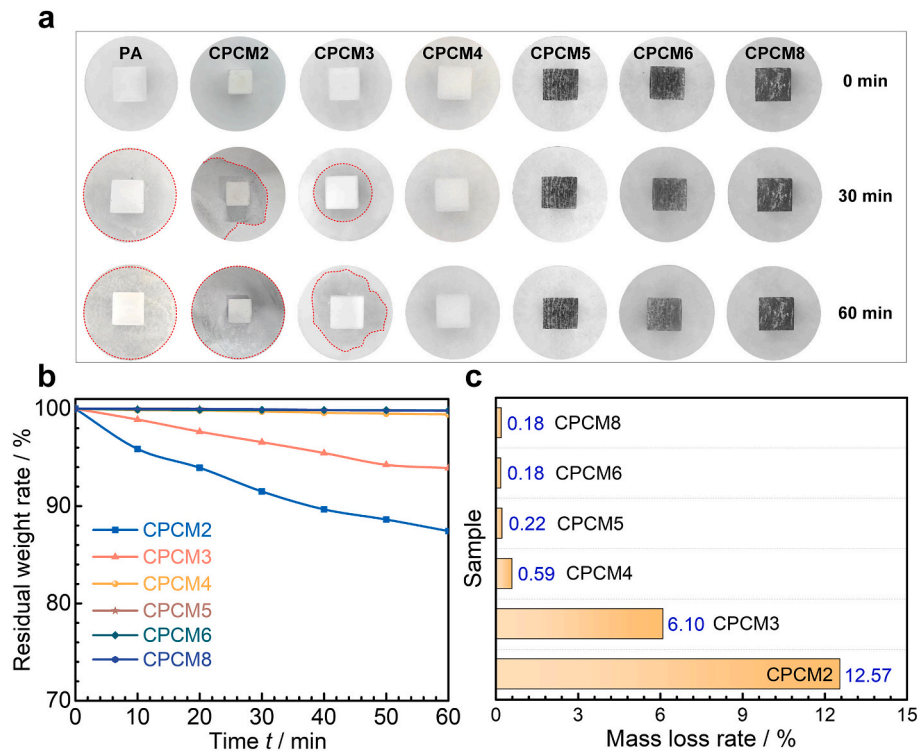


Fig. 4. Leakage-proof properties of the CPCMs. (a) Leakage behaviors for the pure PA and CPCMs at 80 °C for 60 min. (b) Residual weight rates of different CPCMs at different time points. (c) Mass loss rates of the CPCMs.

the CPCMs during the heating/cooling process. To accelerate the heat charging/discharging inside the CPCMs, carbon-based materials are incorporated into the PCMs to boost the thermal conductivity of CPCMs. Nevertheless, current typical CPCMs still exist two problems. One is that low vertical thermal conductivity can cause the temperature of the upper surface of CPCMs to be high, resulting in considerable heat dissipation to the environment. The other is that most CPCMs exhibit isotropic thermal conductivity, and their heat transfer capacities are not well matched with the heat transfer direction from top to bottom. Consequently, more heat is laterally transferred to the surroundings, leading to a decrease in solar-to-thermal efficiency. To address these issues, it is necessary to increase the vertical thermal conductivity for efficient directional thermal transport and decrease the horizontal thermal conductivity of the CPCMs to minimize heat dissipation to the outside.

To realize the above purposes, the CFs with anisotropic thermal conductivity are employed within the PCMs to construct anisotropic and continuous thermally conductive pathways. These pathways can strengthen the heat transfer capacity in the vertical direction, thus increasing the vertical thermal conductivity of the CPCMs effectively. Meanwhile, these pathways also extremely weaken the heat conduction in the horizontal direction, decreasing the horizontal thermal conductivity, and therefore leading to low heat loss. To better investigate the anisotropic heat transfer mechanisms of the CPCMs, the typical series and parallel models are used (Fig. 5). In the two models, heat transfer within the composite is similar to an electric circuit, where thermal resistances are linked either in series or parallel. These thermal resistances are influenced by the thermal conductivity and proportion of the CPCMs. As shown in Fig. 5, the CFs perpendicular to the direction of heat conduction can maximize the thermal resistance of the CPCMs in the series model, which can limit most of the heat transfer inside the CPCMs. Then, in the parallel model, the direction of heat conduction is parallel to the orientation of the CFs, which can minimize the thermal resistance within the CPCMs, thus accelerating the heating/cooling process. These models can be applied to estimate the thermal

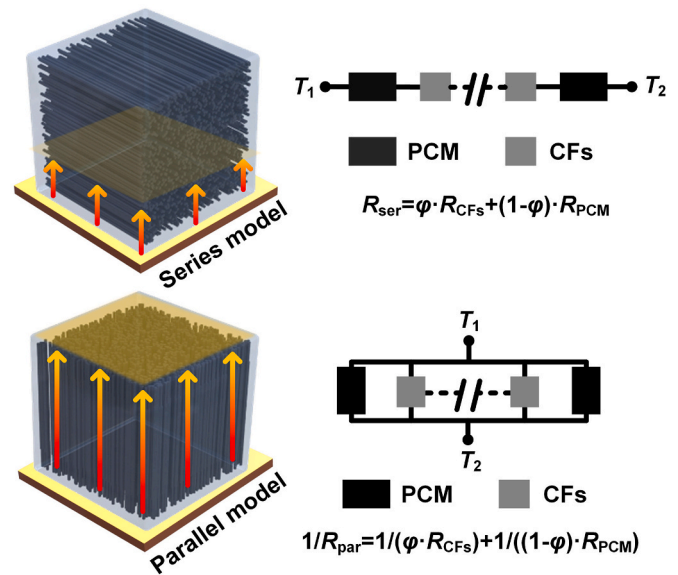


Fig. 5. Schematic of the heat conduction models of the CPCMs.

conductivity of the CPCMs, which can be regarded as a combination of the thermal conductivity of the PA/OBC blends and CFs based on the effective medium theory.

Based on the series and parallel models, the theoretical vertical thermal conductivity of the CPCM8 with 20 wt% CFs was compared with experimental results from Fig. 6b. It can be seen that the theoretical vertical thermal conductivity of the CPCM8 with 20 wt% CFs can be up to 45.19 W·K⁻¹·m⁻¹ (obtained in Note S3-S4). However, the experimental data indicates that its vertical thermal conductivity is only 5.84 W·K⁻¹·m⁻¹, which is much lower than its theoretical value. To investigate the reasons for the large difference between the theoretical and

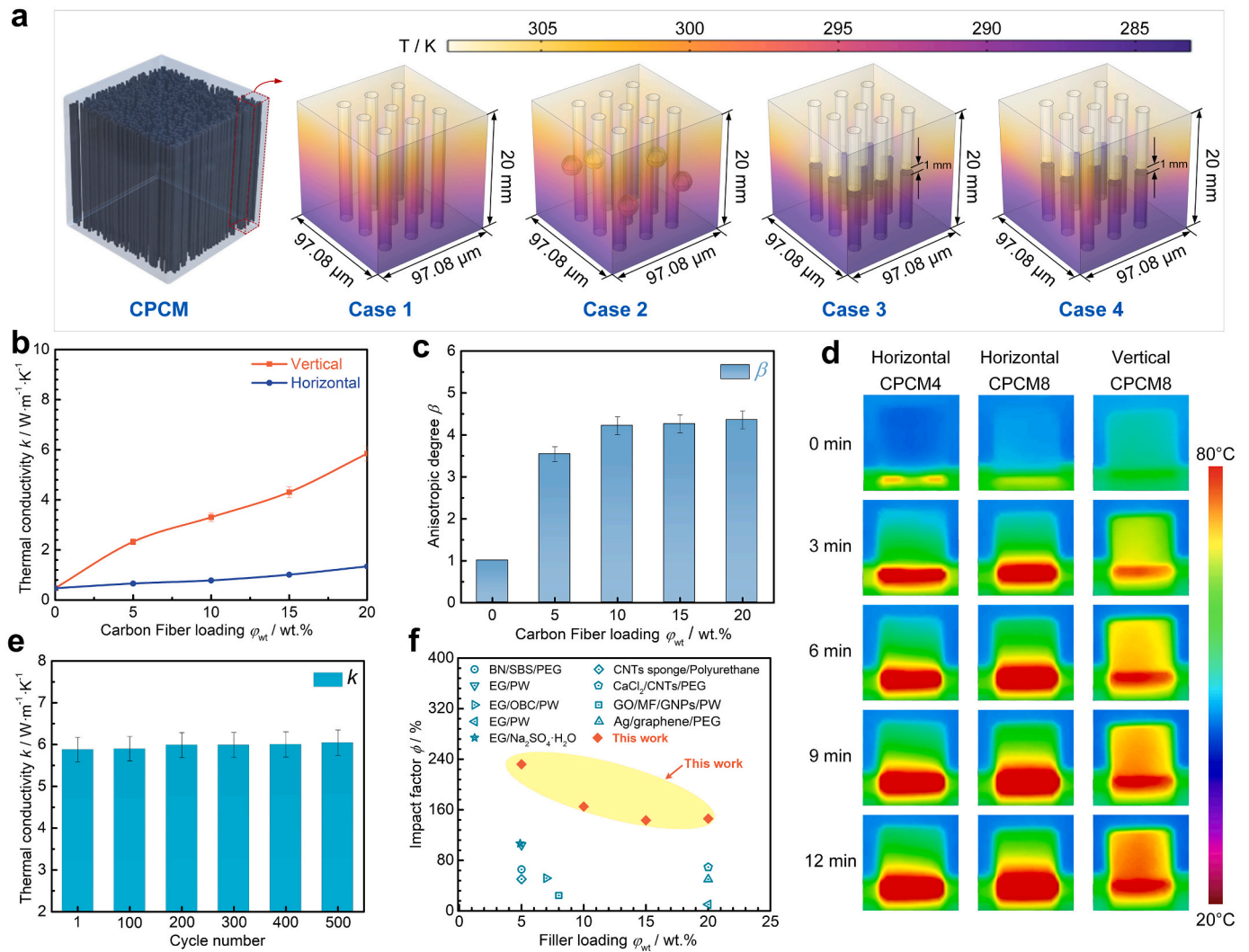


Fig. 6. Anisotropic thermal conductivity and heat transfer characteristics of the CPCMs. (a) Temperature profiles of different cases, where case 1 is nine single CFs that are completely continuous within the composite, case 2 is nine single CFs that are completely continuous, with air bubbles with a volume fraction of 1% attached to their surfaces, case 3 is nine single discontinuous CFs, with a gap of 1 mm in height in the middle of each of them, and case 4 is only one completely continuous CF inside the composite. (b) Vertical and horizontal thermal conductivities of the CPCMs with different CF loadings. (c) Anisotropic degree (β) of the CPCMs at different CF loadings. (d) Infrared images of the CPC4 and CPC8 for heat transfer performance test in both vertical and horizontal directions. (e) Thermal conductivity of the CPC8 during 500 melting-crystallization cycles at 25–80 °C. (f) Comparisons of the impact factors for present CPCMs and the latest reported CPCMs [18,26,43–49].

experimental vertical thermal conductivities of the CPCM, finite element simulation using commercial software COMSOL Multiphysics 6.0 was conducted, and the specific calculated information can be found in Note S5 in the supplementary information. It can be seen that Fig. 6 illustrates the temperature distribution of four different cases, where case 1 is nine single CFs that are completely continuous within the composite, case 2 is nine single CFs that are completely continuous, with air bubbles with a volume fraction of 1% attached to their surfaces, case 3 is nine single discontinuous CFs, with a gap of 1 mm in height in the middle of each of them, and case 4 is only one completely continuous CF inside the composite.

For case 1, it can be found that the temperature within the composite gradually decreases from the top to bottom due to the continuous CFs. Moreover, based on Fourier's Law, the simulated vertical thermal conductivity of the CPC8 with 20 wt% CFs can be calculated as 45.18 W·m⁻¹·K⁻¹ (obtained in Note S5), which is almost the same as the theoretical vertical thermal conductivity. Then, it can be seen in case 2 that there are air bubbles with a volume fraction of 1% attached to the surfaces of the continuous CFs, but the temperature contour of the

composite is almost unchanged compared with that of case 1. Meanwhile, the simulated vertical thermal conductivity also remains around 45.18 W·m⁻¹·K⁻¹, demonstrating that the air bubbles influence a little on the vertical thermal conductivity of the composite. For case 3 and case 4, the temperature contours of the composite vary greatly due to the fracture of the CFs. This is due to the large interfacial thermal resistance generated at the fracture, which hinders the heat transfer inside the composite, thus leading to low thermal conductivity. Besides, the simulated vertical thermal conductivities of the composite in the two cases are 1.15 W·m⁻¹·K⁻¹ and 6.06 W·m⁻¹·K⁻¹, respectively, which are much lower than that of case 1, indicating that the continuity of the CFs influences the thermal conductivity significantly.

In addition, Fig. S2 illustrates the temperature contours of the CPC8 with the CFs with different degrees of fracture. The results show lower thermal conductivity in these cases, with the values remaining at 5.76 W·m⁻¹·K⁻¹–6.06 W·m⁻¹·K⁻¹, which is relatively in line with the experimental vertical thermal conductivity. Meanwhile, it can be further found that the decrease in thermal conductivity depends on the number of fractured CFs and the size of the gap at the fracture location. When the

number of fractures and the size of gaps increase, the thermal resistance at the CF-CF interface can be increased, resulting in a continuously decreasing vertical thermal conductivity. This phenomenon further explains the reasons for the large difference between the theoretical and experimental vertical thermal conductivities. Besides, it also can be seen from SEM images in Fig. 2 that partial CFs within the composite are actually fractured, leading to the low experimental vertical thermal conductivity of the CPCMs.

To sum up, it can be observed that air bubbles have little effect on the vertical thermal conductivity of the CPCMs, while the continuity of the CFs influences it significantly.

Then, the vertical and horizontal thermal conductivities of the CPCMs with different mass fractions of CFs were compared to analyze the heat transfer abilities in two orthotropic directions (Fig. 6b). The vertical and horizontal thermal conductivities rise approximately linearly with increasing CF loading. This means the thermal conductivity enhancement can be readily controlled by adjusting the CF loading. Moreover, the incremental range of the vertical thermal conductivity is much higher than that of its horizontal thermal conductivity. For example, the vertical thermal conductivities of the CPCMs can reach 2.32–5.84 W·m⁻¹·K⁻¹ at CF loadings of 5–20 wt%, and their horizontal thermal conductivities are only enhanced from 0.66 to 1.34 W·m⁻¹·K⁻¹ at the corresponding CF loading. As a result, the anisotropic degree defined as k_V/k_H can be increased from 1.00 to 4.36 (Fig. 6c). Furthermore, it can be observed from Fig. 6c that the anisotropic degree firstly arises rapidly and then rises slightly with increasing CF loading. This is mainly because the excessive CFs can lead to higher interfacial thermal resistance at the CF-CF interface and CF-PCM interface, which can reduce the incremental range of the vertical and horizontal thermal conductivity, limiting the enhancement of the anisotropic degree with increasing CF loading [41].

Then, to illustrate visually the differential heat conduction properties in both vertical and horizontal directions of the CPCMs, the CPCMs and the CPCMs with the same thickness of around 2 cm were put on a heating disk at 80 °C to heat their bottoms, and an infrared camera was used to record the surface temperature distributions of the CPCMs (Fig. 6d). For the CPCMs, one side of the composite is chosen to be heated owing to its isotropic thermal conductivity. As shown in Fig. 6d, the color of infrared images at the bottom of the side in the CPCMs becomes brighter gradually with increasing the heating time, indicating that the bottom surface temperature of the composite rises over time. However, the upper surface temperature is still low throughout the whole heating process. This phenomenon is attributed to the low heating rate caused by isotropic and low thermal conductivity of the CPCMs, which prevents heat conduction from its bottom to its deep inside.

Then, the temperature profiles of the CPCMs in both vertical and horizontal directions were also studied. It can be found that the temperature distributions of the composite exhibit significant variations owing to the high anisotropic thermal conductivity of the composite. When the CFs are parallel to the heating disk, the upper surface temperature of the CPCMs is only up to around 41 °C after heating for 12 min (Fig. 6d), which is similar to that of the CPCMs. However, when the CFs are vertical to the heating disk, the upper surface temperature of the CPCMs can be elevated to 80 °C during the whole heating process. The reason for the significant difference in heat transfer abilities in the horizontal and vertical directions is due to the fact that the CFs within the CPCMs exhibit excellent anisotropic thermal conductivity, where their vertical thermal conductivities are significantly higher than the values in the horizontal direction. When the alignment of the CFs is identical to the direction of the heat transfer, the thermal conductivity of the composite can be obviously enhanced, thus realizing highly efficient thermal transport and fast thermal response.

Furthermore, the vertical thermal conductivity of the CPCMs was tested during 500 melting-crystallization cycles at 25–80 °C (Fig. 6e). It can be found that a slight increase in thermal conductivity after many melting-crystallization cycles, with the value increasing from 5.84 to

6.04 W·K⁻¹·m⁻¹, which is attributed to increased CF loading within the CPCMs due to leakage of partially molten PA after multiple cycles. Meanwhile, the result further demonstrates that the CPCMs can show excellent stability available for long-term efficient thermal energy harvesting.

Besides, to further assess the impacts of the CF loading on the thermal conductivity enhancement of the PCM, an impact factor is defined by Eq. (3) [50,51].

$$\phi = \frac{\lambda_{\text{CPCM}}}{100 \cdot \lambda_{\text{PCM}} \cdot \varphi_{\text{filler}}} \times 100\% \quad (3)$$

where λ_{CPCM} represents the vertical thermal conductivity of the CPCMs; φ_{filler} represents the proportion of the filler; λ_{PCM} represents the vertical thermal conductivity of the PCM; ϕ represents the improved degree of thermal conductivity per 1 wt% filler loading. Compared with the latest reported CPCMs, the impact factor of the present CPCMs can be up to 146% per 1 wt% filler loading (Fig. 6f, Table S3). For the CPCMs with expanded graphite, graphene, graphite, etc., they present relatively low impact factors (<110%). This marked thermal conductivity enhancement of the composites can be ascribed to three reasons. Firstly, the vertical thermal conductivity of the CFs is around 600 W·m⁻¹·K⁻¹, which can improve the thermal conductivity within the composite effectively. Then, oriental CFs can form efficient heat conduction paths within the CPCMs, resulting in accelerated directional thermal transport and rapid thermal response to temperature. Finally, the CPCMs are obtained by compression at 30 MPa during the fabrication process. Its manufacturing method can increase the contact between the CFs and PCMs, decreasing the interfacial thermal resistance, and thus boosting the heat conduction within the CPCMs.

3.5. Solar-to-thermal energy conversion and storage performance

The solar irradiance absorption performance is a crucial factor for achieving CPCMs with high solar-to-thermal efficiency. To assess the absorption performance of the composite, spectral absorptance testing and analysis were performed on the pure PA, CPCMs, and CPCMs with carbon black (Fig. 7a–b). Firstly, the spectral absorptance of the PA varies obviously in the wavelength of 0.3–2.5 μm, and its value can be decreased from 0.920 to 0.44. Meanwhile, the PA also exhibits a low total solar absorptance of 0.558 (obtained in Note S6). For the CPCMs, its spectral absorptance can up to 0.919–0.938 due to the addition of the CFs, resulting in a high total solar absorptance of 0.935 (Fig. 7b). In addition, applying carbon black to the upper surface of the CPCMs can result in high spectral absorptances of 0.944–0.968 when the wavelength range is from 0.3 μm to 2.5 μm, and the total solar absorptance can be as high as 0.966, which is 0.408 and 0.031 higher than that of the PA and CPCMs, respectively (Fig. 7b). The above observation demonstrates that the CPCMs with carbon black can boost the solar irradiance absorption capability effectively.

Fig. 7c illustrates the schematic of an experimental device for assessing the solar-to-thermal energy conversion and storage performance of the CPCMs, including the solar simulator, thermally insulating material, and data recorder. The thermally insulating material is wrapped around the sides and bottom of the composite to decrease heat loss. The thermocouple is placed in the bottom center of the CPCMs for monitoring temperature variations (Fig. S3).

Then, the temperature variations of different CPCMs at 1 kW·m⁻² were measured, as shown in Fig. 7d. It can be seen that the bottom center temperatures of CPCMs have the same trend with time. For these CPCMs, the temperature rises rapidly as the simulated light source turns on, indicating good solar absorption of the samples. Then, slight temperature platforms occur within the temperature of 55–60 °C, suggesting that the phase transitions of the CPCMs appear during this process. Subsequently, the temperature increases rapidly again with the end of the phase transition (Fig. 7d). It can be observed that the phase

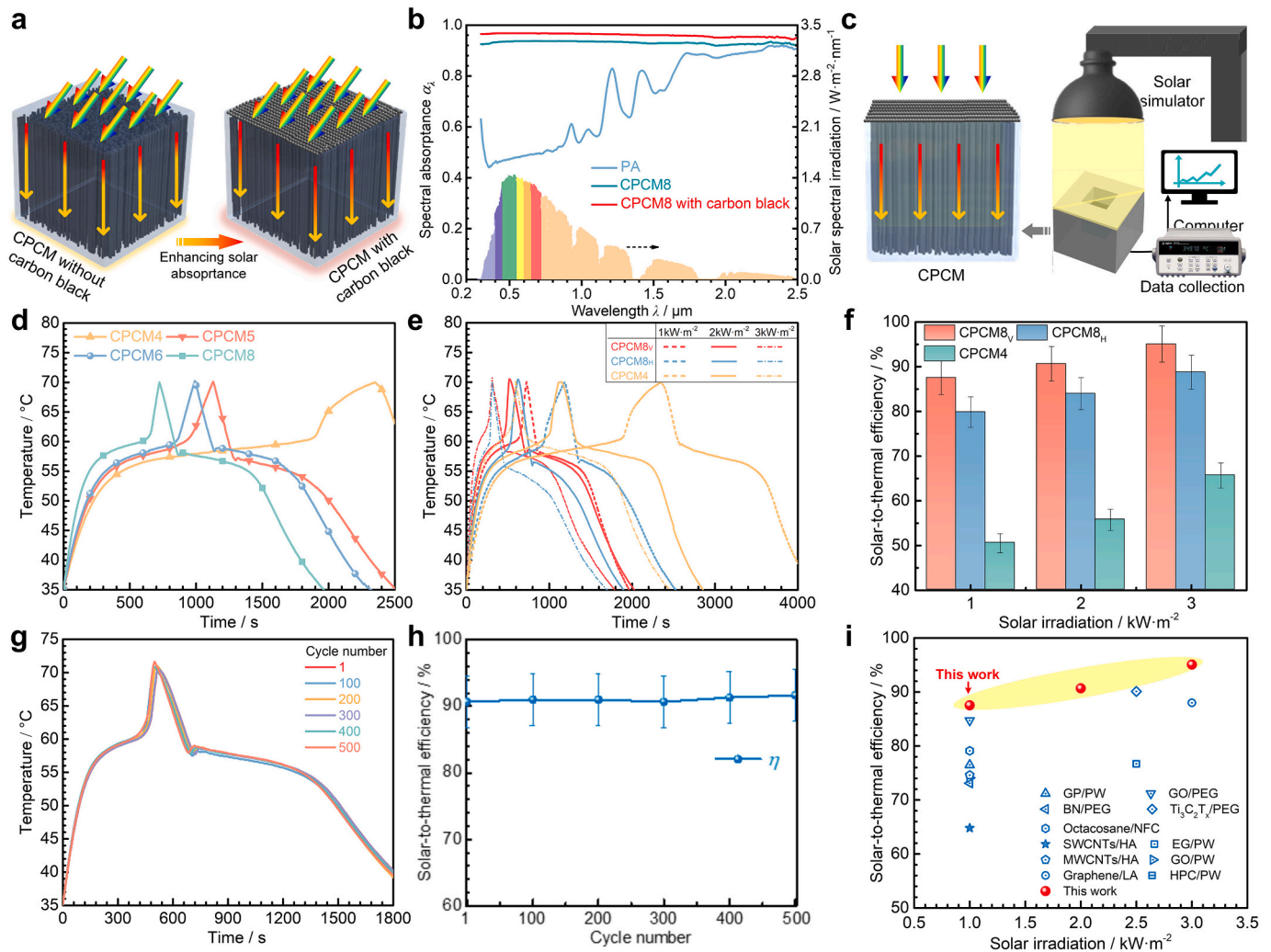


Fig. 7. Solar-to-thermal energy conversion and storage performance of the CPCMs. (a) Schematics of two solar-to-thermal structures. (b) Spectral absorptances of the CPCM8, CPCM8 with carbon black, and pure PA. (c) Schematic of the experimental device to test solar-to-thermal conversion and storage performance. (d) Temperature curves at the bottom centers of the CPCM4–CPCM8 at $1 \text{ kW} \cdot \text{m}^{-2}$. (e) Temperature curves at the bottom centers of the CPCM8_v, CPCM8_H, and CPCM4 at 1 – $3 \text{ kW} \cdot \text{m}^{-2}$. (f) Solar-to-thermal efficiencies of the CPCM8_v, CPCM8_H, and CPCM4 at 1 – $3 \text{ kW} \cdot \text{m}^{-2}$. (g)–(h) Temperature curves at the bottom center and solar-to-thermal efficiencies of the CPCM8_v under 500 heating/cooling cycles at $2 \text{ kW} \cdot \text{m}^{-2}$. (i) Comparisons of the solar-to-thermal efficiencies of the CPCMs with different CF loadings and other latest reported CPCMs [27,52–59].

transition time of the CPCM8 with 20 wt% CFs is 580 s, which is shorter than that of the other composites. Meanwhile, the time point for the CPCM8 to complete the phase transition at the same irradiation intensity is also much shorter than that of the other composites. The above results demonstrate that the heat transfer behavior gets more sensitive and the temperature rises at a faster rate with increasing CF loading, which further indicates that the presence of CFs can efficiently boost the thermal transport in the composite.

Then, the solar-to-thermal conversion and storage experiments of the CPCM4, CPCM8_v, and CPCM8_H were conducted under 1 – $3 \text{ kW} \cdot \text{m}^{-2}$, as shown in Fig. 7e. It can be observed that the phase transition time of each composite decreases with increasing the solar irradiance. For example, when the solar irradiance is enhanced from $1 \text{ kW} \cdot \text{m}^{-2}$ to $2 \text{ kW} \cdot \text{m}^{-2}$, the CPCM4, CPCM8_v, and CPCM8_H can be reduced by 960 s, 303 s, and 427 s, respectively, demonstrating that the increase in the solar irradiance can lead to a shorter phase transition time, and thus contributing to the improvement of solar-to-thermal conversion and storage performance. Furthermore, large differences in the decrease in phase transition time and the increase in temperature can be observed for different CPCMs in the temperature evolutions (Fig. 7e). This is because the high vertical thermal conductivity of the CPCM8_v can

accelerate thermal transport inside the composite, which can result in the fact that the temperature of the CPCM8_v enhances much quicker and the phase transition time becomes shorter than that of the CPCM8_H and CPCM4.

Furthermore, the solar-to-thermal efficiencies of the CPCM4, CPCM8_v, and CPCM8_H were calculated by Eq. (1). It can be observed in Fig. 7f and Table S4–S6 that the solar-to-thermal efficiencies of the CPCM8_v can increase from 87.54% to 95.08% with increasing the solar irradiance from $1 \text{ kW} \cdot \text{m}^{-2}$ to $3 \text{ kW} \cdot \text{m}^{-2}$, which is much higher than those of the CPCM4 and CPCM8_H at the corresponding solar irradiance. This is because elevated solar irradiance can reduce the energy storage time, which can help to reduce heat loss. Meanwhile, the high thermal conductivity in the CPCMs can lead to rapid heat transfer, which can effectively elevate the solar-to-thermal efficiency. In addition, to further assess the cyclic stability of the composite, 500 heating/cooling cycles were performed for the CPCM8 (Fig. 7g). The finding presents that the temperature evolution and the solar-to-thermal efficiency change a little after many repeated cycles (Fig. 7h), confirming that the CPCM-based solar-to-thermal conversion and storage device can exhibit outstanding cyclic stability over extended periods of utilization.

Finally, compared with the latest reported CPCMs, the CPCM8

presents higher solar-to-thermal efficiencies at different solar irradiances (Fig. 7i, Table S7). This is because the high anisotropic thermal conductivity of the composite can accelerate the vertical heat conduction and weaken the horizontal heat conduction inside the composite, which can reduce heat loss to the surroundings, and therefore enhancing the solar-to-thermal efficiency.

The above results demonstrate that the synergetic effects of high vertical thermal conductivity inside the composite and good solar absorption of the carbon black coated at the surface of the composite can achieve efficient solar thermal energy utilization. Besides, since the proposed manufacturing methods are easy to prepare and can be used for scale-up production, the proposed CPCMs could be an ideal option for solar thermal energy conversion and storage.

4. Conclusion

In this work, we proposed a simple method for fabricating anisotropically conductive composite phase change materials (CPCMs) by mixing the palmitic acid (PA)/olefin block copolymer (OBC) mixtures and continuous carbon fibers (CFs) using pressure induction and vacuum treatment. In the obtained CPCMs, CFs contribute to long and continuous heat conduction paths within the composite, leading to high vertical thermal conductivities of $0.47\text{--}5.84\text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and low horizontal thermal conductivities of $0.47\text{--}1.34\text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at CF loadings of 0–20 wt%. Consequently, the anisotropic degree experiences a substantial increase from 1.00 to 4.36. Furthermore, due to the combination of the high solar absorptance of the carbon black and the high vertical thermal conductivity within the composite, the CPCMs exhibit excellent solar-to-thermal efficiencies of $87.54\%\sim 95.08\%$ at $1\text{--}3\text{ kW}\cdot\text{m}^{-2}$. In addition, the CPCMs also exhibit outstanding leakage-proof properties, thermal stability, and cyclic stability for long-term utilization. This work presents an easy and scalable approach for synthesizing high-performance CPCMs for solar thermal energy conversion and storage.

CRediT authorship contribution statement

Pengfei Zhang: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Yilin Wang:** Writing – original draft, Visualization, Investigation. **Yu Qiu:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization. **Hongjie Yan:** Investigation, Formal analysis. **Zhaolong Wang:** Investigation, Formal analysis. **Qing Li:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 52006247, No. 52176093), and the Central South University Innovation-Driven Research Programme (No. 2023CXQD055).

The authors are grateful for computation resources from the High Performance Computing Center of Central South University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2023.122546>.

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